

Chapter 15: Tax Welfare and Optimal Taxation

Public Finance in Canada, 7th edition

Draft

Every tax influences both what governments can do and how people choose to act. When taxes change relative prices, they alter behaviour—discouraging some activities, encouraging others, and redistributing opportunities. These behavioural adjustments create wedges between what buyers pay and what sellers receive, reducing total economic surplus. The resulting deadweight loss represents the efficiency cost of raising revenue—society’s first glimpse of what economists call the Marginal Cost of Public Funds (MCPF).

But efficiency is not the only goal. As we saw in Chapter 3, welfare economics provides a framework for assessing policy not just by how large the economic “pie” is, but also by how it is shared. Traditional models, from *Pareto efficiency* to *Kaldor–Hicks compensation*, treat efficiency and equity as separate domains. Contemporary and intersectional perspectives ([Green et al., 2023](#); [Tedds 2023](#)) challenge that separation, showing that justice and efficiency are intertwined: who bears the cost of taxes and who benefits from public spending depends on power, identity, and access.

This chapter applies welfare economics directly to tax design. Building on Chapter 14’s focus on who pays, we now ask how taxation affects total welfare and how efficiency and equity can be reconciled in an optimal tax system. We begin by showing how taxes reduce economic surplus and create deadweight loss—the foundation of the *Harberger wedge*—and then derive how elasticities and tax rates determine the magnitude of that loss. We next link efficiency and distributional concerns through the *Marginal Cost of Public Funds (MCPF)* and the *Marginal Value of Public Funds (MVPF)*—applied welfare measures that quantify how costly or valuable a tax-financed dollar can be.

Taken together, these tools form a unified framework for evaluating tax policy and public spending—not only within markets but across society. The chapter concludes with the optimal-tax framework, which integrates efficiency, equity, and behavioural response to guide real-world fiscal design. Throughout, we emphasize that efficiency is not value-neutral: an “efficient” system that excludes or burdens some groups is socially wasteful. A truly efficient and equitable system raises the resources needed for collective well-being, reflects the diversity of those it serves, and provides a disciplined way to think about choices among tax cuts, public spending, and debt financing.

WHY EFFICIENCY MATTERS

Taxes make collective life possible. They finance the public goods and services that sustain communities and create opportunities. Yet every tax also changes the way people and firms behave. When a tax alters prices, wages, or profits, it shifts incentives and prevents some mutually beneficial trades from occurring. The value of those foregone trades represents a loss of economic efficiency—a reduction in total welfare that benefits no one.

Economists visualize this change as a wedge between the price buyers pay and the price sellers receive—the Harberger wedge. This wedge captures how a tax distorts market outcomes and becomes the building block for the modern concept of the Marginal Cost of Public Funds (MCPF).

From Welfare Economics to Taxation

As discussed in Chapter 3, welfare economics provides the benchmark for evaluating how policies affect overall well-being. There we learned that total welfare in a market can be represented by economic surplus, the sum of consumer surplus and producer surplus. A policy change that increases total surplus is a potential Pareto improvement; one that reduces it is welfare-reducing.

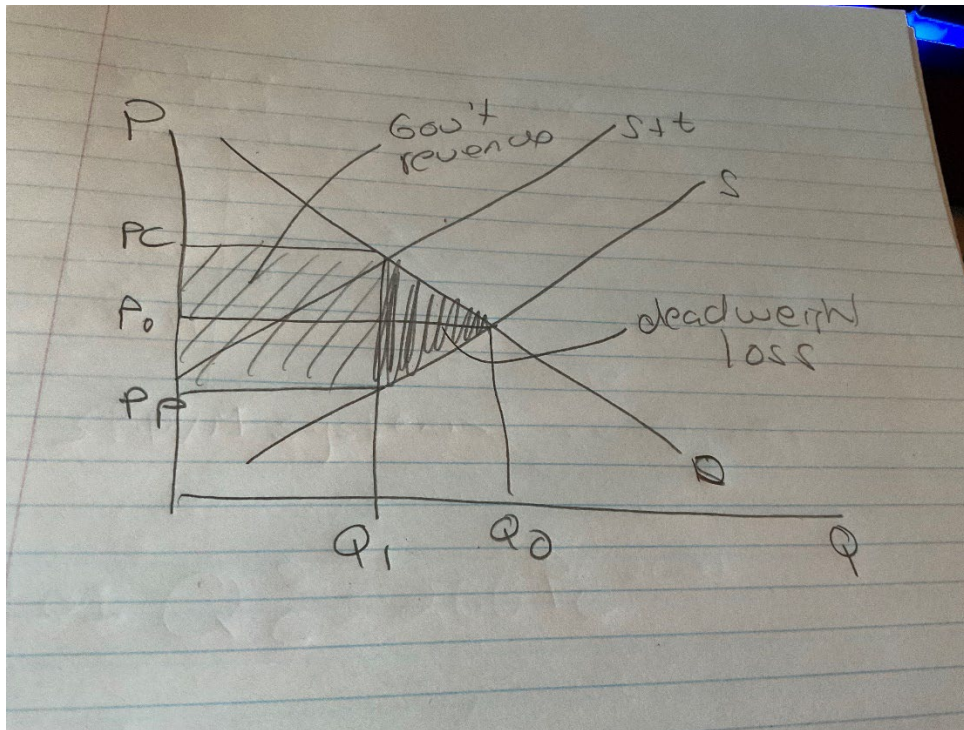
Here, we apply that same reasoning to taxation itself. A tax collects part of that surplus for public use but also destroys part of it through behavioural adjustment.

Taxes and the Loss of Surplus

Consider a competitive market for a good—say bicycles as shown in Figure 15.1. Before taxation, equilibrium price and quantity are P_0 and Q_0 . Suppose government imposes a per-unit tax t . The price paid by buyers rises to P_c , the price received by sellers falls to P_p , and quantity traded falls to Q_1 .

The vertical gap $t = P_c - P_p$ is the Harberger wedge. The rectangle between P_c and P_p up to Q_1 represents government revenue—a transfer from private agents to the public sector. The small triangle between Q_1 and Q_0 represents the deadweight loss (DWL): surplus that disappears because the tax prevents some trades that would have benefited both buyers and sellers.

FIGURE ## Deadweight loss of taxation



Deriving the Efficiency Loss

We can approximate this welfare loss formally.

The area of the lost triangle equals

$$DWL = \frac{1}{2}(P_c - P_p)(Q_0 - Q_1) = \frac{1}{2}\Delta Q.$$

Because the quantity change $\Delta Q/Q_0$ is determined by the relevant elasticity (η) and the proportional price change t/P ,

$$\frac{\Delta Q}{Q_0} = \eta \frac{t}{P_0},$$

we can express the loss as

$$DWL \approx \frac{1}{2}Q_0P_0\eta\left(\frac{t}{P_0}\right)^2.$$

This expression reveals the geometry behind the Harberger triangle and introduces the logic of the MCPF: as the tax wedge grows, the additional efficiency cost of each revenue dollar rises.

Two core insights follow.

1. First, the efficiency cost increases with the elasticity of demand or supply: the more responsive behaviour is to price, the larger the reduction in traded quantity.
2. Second, the cost rises with the square of the tax rate—doubling the rate roughly quadruples the loss.

Broad tax bases with low rates therefore raise revenue more efficiently than narrow, high-rate taxes.

Interpreting the Welfare Triangle

The *deadweight loss* represents lost potential *Pareto improvements*: trades that would have left both buyers and sellers better off without harming anyone else. It is not money that transfers from one group to another but welfare that vanishes from the economy. Governments accept this loss in order to raise revenue for public purposes, but the aim is to minimize it per dollar collected.

This is precisely the motivation behind the MCPF: to measure how much social cost is incurred for each extra dollar of revenue. When the wedge is small, the marginal cost of funds is close to one; when the wedge widens, that cost rises.

Because the ability to adjust differs across people and sectors, these efficiency losses also reflect social and institutional structure. Workers with inflexible schedules, small firms bound by contracts, or residents of remote or northern communities often have more inelastic responses and therefore bear heavier burdens. Indigenous governments operating under unique fiscal and geographic constraints may face higher administrative costs when shifting tax bases. Efficiency, in other words, is not purely mechanical—it depends on who can adapt and who cannot.

Efficiency, Fairness, and Design

All taxes distort behaviour, but not equally. Taxes on goods or factors that are less elastic—such as urban land or resource rents—create smaller efficiency costs. Taxes on highly elastic bases—like mobile capital or certain forms of labour—create larger losses. An efficient and equitable system therefore raises required revenue where behaviour changes least, while using spending and credits to maintain fairness.

From Welfare Effects to Measurement

Efficiency analysis links taxation directly to welfare economics. Taxes shrink total economic surplus because they change behaviour, and that loss depends on how responsive people and firms are to price changes. The Harberger wedge translates this behavioural response into a measurable welfare cost, and the MCPF formalizes it. Understanding these effects is the foundation for evaluating and designing tax systems that raise revenue with minimal distortion and reflect society's commitments to fairness and inclusion.

The next section quantifies these effects more precisely and shows how elasticities and tax rates determine the magnitude of the efficiency cost of taxation—and, ultimately, the marginal cost of public funds.

MEASURING THE EFFICIENCY COST OF A TAX

The previous section showed that every tax reduces total economic surplus by discouraging some mutually beneficial trades. In this section, we quantify that welfare loss, identify the behavioural factors that determine its size, and connect these relationships to the concepts of the Marginal Cost of Public Funds. Measuring efficiency loss turns a qualitative concept—“some value is lost”—into a quantitative one that can inform tax policy.

Elasticities and the Shape of the Loss

The responsiveness of buyers and sellers determines how far quantity falls when a tax is imposed. The greater this response, the larger the deadweight loss (DWL). As shown in Section 15.1, if demand and supply are roughly linear near the equilibrium, the welfare loss can be expressed as:

$$DWL \approx \frac{1}{2} Q_0 P_0 \eta \left(\frac{t}{P_0} \right)^2.$$

where Q_0 and P_0 are the pre-tax equilibrium quantity and price, t is the per-unit tax, and η represents the relevant elasticity. This equation visualizes the “Harberger triangle” in algebraic form and makes clear that the efficiency cost grows faster than revenue as the wedge widens. The relationship reveals two important insights:

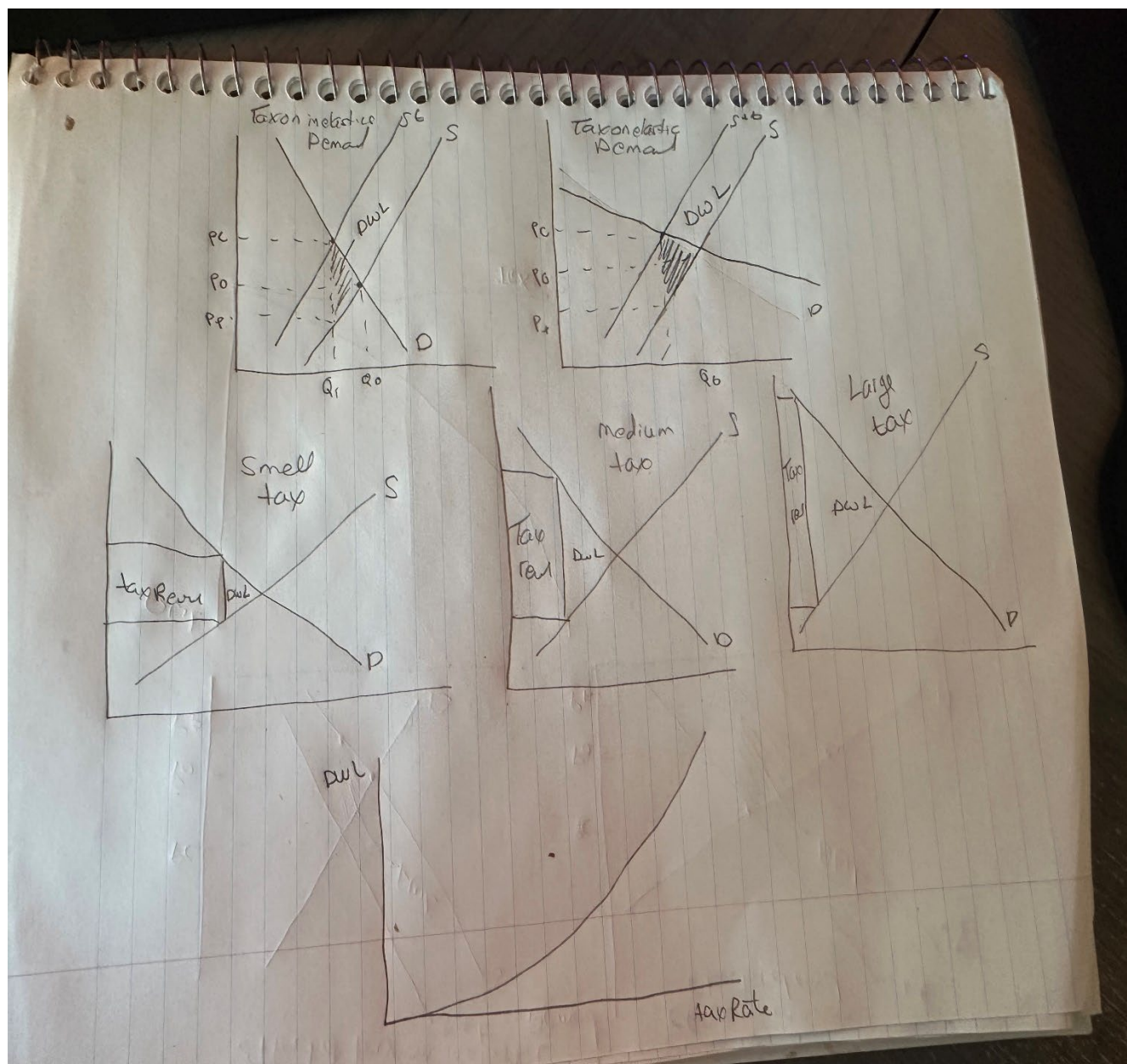
1. Elasticities matter. When consumers or producers are highly responsive to price, they reduce quantity substantially, increasing the efficiency loss.
2. Tax rates matter even more. Because the loss rises with the square of the tax rate, doubling a rate roughly quadruples the distortion.

Figure 15.2 summarizes the key determinants of the efficiency cost of a tax. Panel (a) compares markets with different elasticities: inelastic markets (with steep demand and supply) experience small deadweight losses, while elastic markets experience large ones. Panel (b) shows that, even holding elasticities constant, losses rise disproportionately with the tax rate—the efficiency cost increases with the square of the rate. Panel (c) extends this logic to tax-base breadth: a narrow, high-rate tax creates concentrated distortion, whereas a broad-base, low-rate tax spreads a smaller wedge across many transactions.

Together, these relationships explain why economists advocate broad-base, low-rate tax structures that raise revenue efficiently and predictably.

These properties explain why economists often favour *broad-base, low-rate tax* systems: spreading revenue collection over many transactions with small wedges minimizes total loss.

FIGURE ## The Determinants of deadweight loss (the deadweight loss in the top left panel should be smaller than the dwl in the top right panel)



Panel (a): Inelastic vs. elastic markets—deadweight loss grows with behavioural responsiveness. Panel (b): For any elasticity, loss rises with the square of the tax rate. Panel (c): A broad tax base spreads distortion thinly, while a narrow, high-rate tax concentrates it.

From the Harberger Triangle to the MCPF

The Harberger wedge shows the average efficiency cost of a tax; the Marginal Cost of Public Funds (MCPF) extends this logic to the margin. It asks: How much additional social cost arises when government raises one more dollar of revenue?

Formally,

$$MCPF = 1 + \frac{\Delta DWL}{\Delta R},$$

When the wedge is small, $\Delta DWL/\Delta R$ is close to zero and $MCPF \approx 1$; each additional dollar of revenue costs society roughly one dollar in lost welfare. As the wedge widens—through higher rates or more elastic behaviour—the MCPF rises above one, meaning every new revenue dollar costs more than it yields. This provides a direct behavioural foundation for thinking about efficient tax design: minimize the welfare cost per dollar raised.

The Role of Tax Bases and Behavioural Margins

Efficiency costs also depend on how many behavioural margins a tax touches. A narrow base—for example, taxing only luxury cars—creates large distortions in a small sector, while leaving other, untaxed goods unaffected. A broad base—such as a value-added tax—applies the same wedge across many goods, reducing opportunities to substitute toward untaxed items and therefore lowering total DWL.

This is why Canada's shift from the federal Manufacturers' Sales Tax to the Goods and Services Tax (GST) in 1991 improved efficiency: it broadened the base and lowered average rates, reducing the overall MCPF.

Substitution and Income Effects

Not all behavioural change represents a true efficiency loss.

A tax affects welfare through two channels:

1. Substitution effect – people substitute away from the taxed good because its relative price rises. These are the changes that generate deadweight loss.
2. Income effect – people feel poorer and may change consumption even if relative prices are constant. These adjustments do not create additional distortion; they merely reflect reduced purchasing power.

Economists separate these effects using *compensated demand* (holding utility constant) and *uncompensated demand* (observed in practice). The classic *Harberger triangle* captures only the substitution effect. Modern microsimulation and computable general-equilibrium (CGE) models extend this logic across many goods and households, producing economy-wide estimates of the MCPF.

Short- and Long-run Elasticities

The efficiency cost of a tax also depends on time. In the short run, consumers and producers may find it difficult to adjust; elasticities are low and efficiency losses small. Over time, they adapt—investing in new equipment, relocating production, or changing work patterns. Long-run elasticities are therefore higher, and long-run deadweight losses larger.

For example, immediately after a fuel-excite tax, commuters have few alternatives; quantity falls little and DWL is modest. Over several years, households may adopt electric vehicles or move closer to work, increasing behavioural response and thus the MCPF. This time dimension reminds policymakers that static efficiency estimates understate the long-run cost of persistent distortions.

Table 15.1 reports approximate Canadian elasticity estimates drawn from recent studies. These values, though stylized, show how responsiveness—and hence efficiency cost—varies across markets.

Table #.#

Empirical Estimate in Context

Tax Base (Market)	Elasticity Range	xxxxx Sources	Placeholder Text
Gasoline (short-run)	-0.2-(-0.3)	xxxxx Dahl and Sterner (1991); Huges, Knittel, & Sperling (2008); and Rivers & Schaufele (2015)	xxxxx
Gasonline (long-run)	-0.5-(-0.8)	xxxxx Dahl and Sterner (1991); Huges, Knittel, & Sperling (2008); and Rivers & Schaufele (2015)	xxxxx
Alcohol	-0.3-(-0.6)	xxxxx Fogarty (2010); Meng, Brennan & Holmes (2014)	xxxxx
Labour supply (Primary earners)	0.1-0.3	xxxxx Bargain, Orsini, & Peichl (2014)	xxxxx
Labour supply (Secondary earners)	0.6-1.0	xxxxx Heim (2007); Eissa & Liebman (1996); Crossley & Jeon (2007)	xxxxx
Corporate Capital (Investment/capital mobility)	>1.0	xxxxx Devereux & Griffith (2003); Mintz & Smart (2004); Bazel & Mintz (2016)	xxxxx
Urban Land	0	xxxxx Arnott & Stiglitz (1979); Tideman (1999); Slack (2011)	xxxxx

Place Citation information here.

The contrast between labour-market elasticities highlights an important equity dimension: workers who can adjust hours easily (often higher-income or secondary earners) bear less burden, while those in inflexible jobs or caregiving roles bear more. This is one reason why efficiency and fairness cannot be separated.

Efficiency, Design, and the Path to the MCPF Measurement

By quantifying efficiency losses, we move from intuition to evaluation. The deadweight-loss formula provides the foundation for the marginal cost of public funds, which formalizes how costly it is to raise one additional dollar through a given tax.

The next section introduces the full MCPF framework and its empirical estimation, connecting these micro-behavioural insights to the welfare metrics that guide modern public-finance design.

Efficiency and Equity in Welfare Economics

Taxes do more than raise revenue—they reshape the distribution of resources and opportunities. Every policy that changes who pays and who benefits implicates both efficiency and equity. Classical welfare economics treated these as separate dimensions: efficiency as the size of the economic pie, and equity as how that pie is divided. Modern welfare analysis—and especially the intersectional approaches introduced in Chapter 3—rejects this separation. Justice and efficiency are not trade-offs to be balanced after the fact; they are jointly determined by how policy affects participation, opportunity, and well-being.

From MCPF to Social Welfare

In the previous section, we defined the Marginal Cost of Public Funds (MCPF) as the efficiency cost of raising one additional dollar of revenue. That measure captures what is lost when taxes distort behaviour. But it tells us nothing about what is gained when those funds are spent.

To complete the welfare calculus, we need a measure of the social value of public spending. The next section introduces the Marginal Value of Public Funds (MVPF)—the welfare gain from an additional dollar of government expenditure. Together, MCPF and MVPF translate the abstract notions of efficiency and equity into operational metrics.

Efficiency and the Social Welfare Function

In welfare economics, the total well-being of society can be represented by a social welfare function (SWF) that combines the utilities of individuals, often denoted as

$$W=W(U_1, U_2, \dots, U_n).$$

The simplest approach—utilitarianism—adds up everyone's utility; other forms weight the utilities of lower-income individuals more heavily, reflecting concern for distribution. Efficiency enters because total welfare rises when output increases, but equity enters because marginal welfare gains are greater for those with lower income.

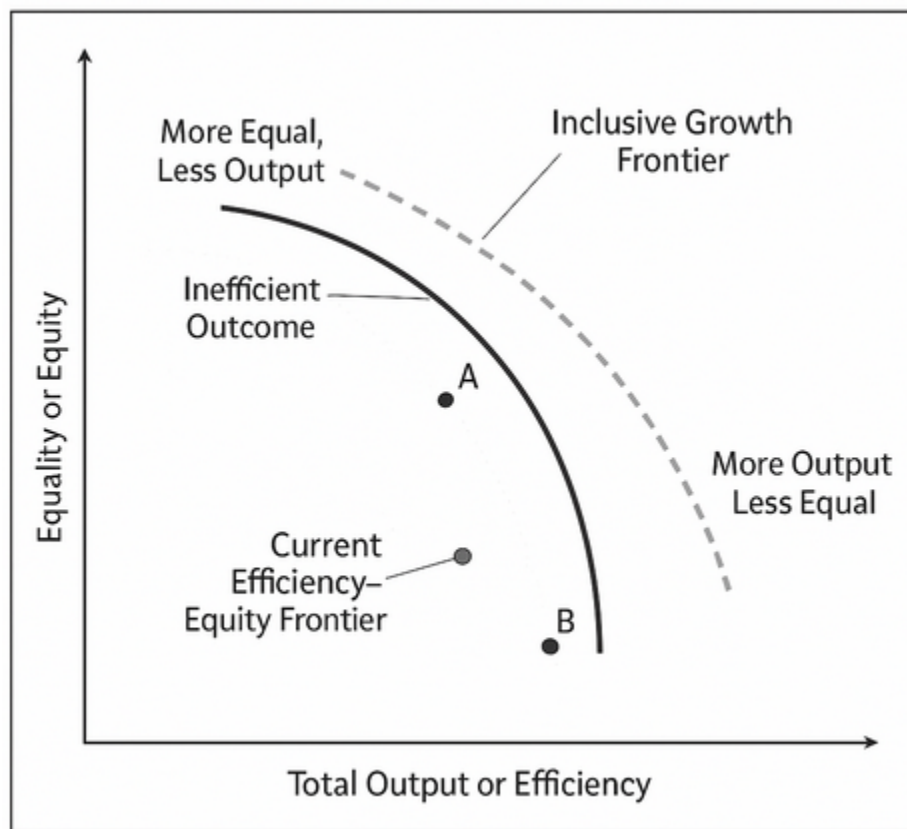
A perfectly Pareto-efficient allocation maximizes total surplus given current resources, but may still be highly unequal. The *Kaldor–Hicks criterion* allows efficiency-improving policies that could make some people worse off, so long as the gains are large enough that winners could compensate losers. Taxation forces us to navigate both criteria at once: a tax may reduce total surplus (an efficiency cost) but increase welfare overall if it funds transfers or services that benefit those with higher marginal utility of income.

The Efficiency-Equity Frontier

Just as firms face a trade-off between output and cost, societies face a trade-off between efficiency and equity. Figure 15.3 illustrates the relationship between total economic output and equality—the efficiency–equity frontier. Points inside the curve represent outcomes that are inefficient: society could achieve greater equality or higher output, or both. Points along the frontier show efficient combinations where improving one goal requires sacrificing the other.

Policies such as redistribution, labour-market reforms, or environmental levies move society along or, in some cases, shift the entire frontier outward by reducing structural barriers and improving participation. Inclusive growth, in this sense, is efficiency-enhancing as well as fair.

FIGURE ## Figure Caption Title



The Classical and Modern View

In the mid-twentieth century, economists such as Arthur Okun and James Mirrlees formalized this trade-off. Okun ([1975](#)) described redistribution as “the leaky bucket”—transfers promote equality but lose some resources to administrative cost and behavioural response. Mirrlees ([1971](#)) later developed the optimal income tax framework, showing how marginal tax rates can be set to balance equity goals against efficiency losses from reduced labour supply.

Building on these insights, Joseph Stiglitz ([2012](#), [2015](#)) advanced the concept of inclusive growth, arguing that equity and efficiency are not opposing ends of a spectrum but mutually

reinforcing goals. Persistent inequality, he showed, undermines productivity, innovation, and democratic legitimacy, while more equitable access to education, credit, and opportunity strengthens overall economic performance. Together, these ideas underpin modern tax design: efficiency and fairness must be pursued jointly if growth is to be both sustainable and broadly shared.

When Efficiency and Equity Align

Although the two objectives often conflict, they can also reinforce each other. Policies that reduce discrimination, improve access to education, or enhance participation in markets can increase both fairness and total efficiency by allowing people to contribute according to their potential. For example, reducing barriers to labour-market entry for women, newcomers, or Indigenous workers expands the tax base, lowers the MCPF, and increase the MVPF of public investment in inclusion. Similarly, efficient environmental taxes that recycle revenue to low-income households can improve both environmental and social outcomes. These are double-dividend policies—where justice enhances efficiency.

An Intersectional Critique

As discussed in Chapter 3, traditional welfare economics assumes that individuals are identical in preferences and constraints except for income. In practice, opportunities and capacities differ across lines of gender, race, ability, and geography. A tax that seems neutral on paper can impose unequal burdens because groups face different elasticities or barriers to adjustment.

For instance, payroll taxes may disproportionately affect workers with caregiving responsibilities, and consumption taxes on energy or food can weigh more heavily on northern and Indigenous communities where substitutes are limited. Recognizing these differences broadens the concept of efficiency itself: a tax system that systematically excludes or disadvantages groups is inefficient in realizing society's potential output.

Modern analyses therefore incorporate distributionally weighted welfare functions, behavioural heterogeneity, and the lived realities of policy participants. This inclusive approach connects economic modelling to the practical pursuit of Equity, Diversity, Inclusion, and Accessibility (EDIA)—a perspective now embedded in many Canadian fiscal frameworks.

Policy Implications

Understanding the efficiency–equity relationship helps policymakers decide which taxes to use and how to combine them with transfers. Taxes on inelastic bases (such as land or resource rents) are efficient but often fall on high-wealth households, aligning efficiency and equity. Taxes on labour or consumption can be efficient when rates are broad and low, but may need credits or benefits to maintain fairness. Targeted transfers financed by mild efficiency costs can raise overall social welfare when the marginal value of income is high for recipients.

Ultimately, the task of tax design is not to minimize efficiency costs at all costs, but to ensure that each dollar of distortion buys as much social benefit as possible.

From Equity and Efficiency to Public Value

Efficiency and equity cannot be separated in practice. The challenge for public finance is to weigh them systematically in fiscal design. The next section introduces the Marginal Value of Public Funds (MVPF)—a tool that formalizes this trade-off by translating efficiency losses and equity gains into a common welfare metric.

Taken together, the MCPF and MVPF create a two-sided measure of public value: one quantifies how much welfare is lost when we raise funds; the other, how much is gained when we spend them. This pairing converts the abstract efficiency–equity frontier into a practical test for public policy design.

From Equity and Efficiency to Public Value

Public finance is about choices: how much governments should spend, how they should raise it, and who ultimately benefits.

Economists have long sought ways to measure these trade-offs in consistent welfare terms. Two complementary tools—the Marginal Cost of Public Funds (MCPF) and the Marginal Value of Public Funds (MVPF)—translate the abstract efficiency–equity frontier into operational measures of public value. Each quantifies, in different ways, how much social welfare changes when one more dollar is raised or spent.

The Logic of the Marginal Cost of Public Funds

Every tax that raises revenue imposes an efficiency cost. The Marginal Cost of Public Funds (MCPF) captures how large that cost is at the margin—how much total welfare society loses for each additional dollar of revenue raised.

MCPF builds directly on the Harberger wedge introduced earlier. When a small change in a tax rate t raises additional revenue, ΔR or Marginal Revenue Collected, and increases ΔDWL or Marginal Deadweight Loss, the marginal cost of funds is:

$$MCPF = 1 + \frac{\text{Marginal Deadweight Loss}}{\text{Marginal Revenue Collected}}$$

An MCPF greater than 1 implies that raising an additional dollar costs society more than a dollar in lost welfare. For example, if the MCPF for the personal income tax is 1.3, then collecting one dollar of extra revenue imposes a social cost of \$1.30 once distortion is considered. The higher the elasticity of the taxed base, the larger the MCPF. Taxes on inelastic bases—urban land, natural resource rents, or pollution—have lower MCPFs because behaviour changes little.

Determinant of the MCPF

1. Elasticities of demand and supply: the greater the behaviour response, the higher the marginal deadweight loss for a given revenue gain.
2. Tax rates and base breadth: narrow, high-rate of taxes raise the MCPF more quickly than broad, low-rate taxes.

3. Complementary institutions: efficiency administration, strong compliance, and coordination across government reduce the MCPF by shrinking compliance costs and behavioural leakages.

Table 15.2 reports stylized estimates from Canadian and international studies, showing how MCPF varies across instruments. Empirical estimates confirm that not all revenue sources impose equal social costs. Taxes on immobile or inelastic bases—such as land, natural resources, or broad consumption—tend to have MCPFs near 1, while those on highly elastic bases, such as mobile corporate capital, can exceed 1.3.

Table #.#

Stylized Estimates of the Marginal Cost of Public Funds

	Approximate MCPF Range	Source	Placeholder Text	Placeholder Text
Land or Resource Rent Taxation	1.00-1.05	Arnott & Stiglitz (1979); Dahlby (2008)	xxxxx	xxxxx
Broad-based Consumption Taxes (e.g., GST)	1.05-1.20	Smart (2011); Ruggeri (1999)	xxxxx	xxxxx
Payroll Taxes	1.10-1.30	Dahlby & Ferede (2012); OECD (2015)	xxxxx	xxxxx
Personal Income Taxes	1.20-1.40	Ballard (1990); Dahlby (2008)	xxxxx	xxxxx
Corporate Income Taxes	1.30-1.80	Bazel & Mintz (2016); Dahlby (2020)	xxxxx	xxxxx

Place Citation information here.

Interpretation and Policy Use

The MCPF helps governments evaluate how to raise revenue, but not whether doing so is worthwhile. A high MCPF warns that further reliance on a given base is inefficient; a low MCPF signals room to expand. Yet an MCPF of 1.25 does not imply taxes should be cut—only that each additional dollar raised costs \$1.25 unless it finances something worth more. However, it is not without its limits.

Critiques and Limitations of the MCPF Framework

While the MCPF is a foundational concept in applied public finance, its interpretation requires caution. It is often presented as a mechanical efficiency measure, yet in practice it embeds several strong assumptions about how governments, markets, and people behave.

1. **Narrow Welfare Scope:** The MCPF captures the efficiency cost of raising funds but says nothing about what those funds accomplish. It assumes that the welfare loss from taxation can be evaluated independently of the welfare gain from spending. In reality, governments raise taxes to finance health care, education, income supports, and environmental protection—each of which generates social returns that may far exceed the measured cost of collection.
2. **Distributional Blind Spots:** The MCPF aggregates welfare losses across individuals without regard to who bears them. A tax that slightly reduces aggregate efficiency but primarily affects high-income households may raise overall welfare once equity weights are applied. Conversely, a low-MCPF tax that falls heavily on low-income groups can still reduce social welfare. In short, a “low” MCPF does not necessarily mean a better tax from a justice perspective.
3. **Static Behaviour:** Standard MCPF estimates assume fixed elasticities and instantaneous adjustment. They typically ignore dynamic feedbacks—such as how tax-funded infrastructure can raise productivity, or how public investment in early childhood education expands future tax bases. These long-run effects can lower the true cost of public funds well below static estimates.
4. **Administrative and Compliance Costs:** Most MCPF calculations neglect the resources required to administer and comply with taxes—costs that vary widely across instruments. A theoretically efficient tax that is costly to enforce, prone to avoidance, or regressive in its administration may have a much higher effective MCPF than the model implies.
5. **Implicit Benchmark of the Status Quo:** The MCPF is a marginal measure—it compares small changes to an existing system. When the system itself is inefficient or inequitable, marginal analysis can obscure opportunities for reform that move the entire tax structure closer to an equitable optimum.

Taken together, these critiques suggest that MCPF is best understood as a partial measure of fiscal efficiency, not a complete metric of social welfare. It helps identify where behavioural distortions are large but must be interpreted alongside distributional, administrative, and long-term effects. When used without these caveats, it risks privileging narrow efficiency gains over broader measures of fairness and social progress. Yet the MCPF captures only the cost side of fiscal policy. To understand whether those costs are justified, we need a parallel measure that values the benefits of spending or transfers—the MVPF.

The Marginal Value of Public Funds (MVPF)

While the MCPF focuses on the cost side of fiscal policy, the MVPF focuses on the benefit side—the welfare return generated by an additional dollar of public spending or tax expenditure.

Developed by Hendren and Sprung-Keyser (2020), the MVPF provides a unifying framework for evaluating diverse government programs—from student grants and childcare subsidies to disability supports and infrastructure investment—on a comparable welfare scale.

$$MVPF = \frac{\text{Social Benefits from a Policy}}{\text{Social Cost of Raising a Dollar}} = \frac{\text{Welfare Gain to Recipients}}{\text{Net Fiscal Cost to Government}}$$

An MVPF of 1 means each dollar of spending yields one dollar of social value. If MVPF is greater than 1, it indicates that the program increases total welfare: every dollar spent yields more than one dollar in social benefit once behavioural and fiscal feedbacks are accounted for. An MVPF less than 1 suggests that the program reduces welfare because its benefits are smaller than its cost of financing and the funds could create greater value elsewhere.

In practical terms:

- MVPF = 1 → the program is welfare-neutral (social benefits equal social costs);
- MVPF > 1 → welfare-improving (e.g., effective education or targeted transfer programs);
- MVPF < 1 → welfare-reducing (e.g., regressive or distortionary subsidies).

The numerator—the welfare gain—is ideally the sum of individuals' willingness to pay (WTP) for the policy. For cash transfers, WTP is close to the amount transferred, especially for liquidity-constrained households. For in-kind programs such as child-care subsidies or public insurance, WTP must be inferred from behaviour: how much people are willing to give up, work, or relocate to obtain the benefit. However, due to ongoing limitations that will be discussed below, in practice, the MVPF is an empirical approximation of the welfare measure of public spending.

Empirical Estimation and Evidence

MVPF analysis has transformed the empirical study of public finance by connecting microeconomic program evaluation to welfare theory. Using data on elasticities, income responses, and long-term outcomes, economists estimate both the fiscal externality (how a program affects government revenue) and the private benefit (how it affects individual welfare).

Programs that raise lifetime earnings, improve health, or expand labour-force participation—such as early-childhood education, job training, or targeted income support—often exhibit MVPF > 1.

Table 15.3 reports stylized estimates from Canadian and international studies, showing how MVPF varies across programs. For example:

- Early-childhood education programs in the U.S. and Canada often have MVPFs between 4 and 10, reflecting large long-run benefits relative to cost.
- Tax credits that primarily accrue to high-income households often have MVPFs between 0.3 and 0.8, indicating net welfare loss.

These comparisons allow policymakers to prioritize interventions that deliver the greatest welfare “return per public dollar.”

Table #.#

Illustrative Estimates from Recent Studies

Program	Approximate MVPF Range	Source	Placeholder Text	Placeholder Text
Early-childhood education and preschool programs	4-10	Hendren & Sprung-Keyser (2020); Heckman et al. (2010)	xxxxx	xxxxx
School meal programs in developed economies	2-5	Card, de Chaisemartin, and Gurgand (2024)	xxxxx	xxxxx
Postsecondary education and student aid	1.5-3	Hendren & Sprung-Keyser (2020); Zimmerman (2014)	xxxxx	xxxxx
Broad untargeted tax expenditures	0.3-0.8	Hendren & Sprung-Keyser (2020); OECD (2021)	xxxxx	xxxxx

Place Citation information here.

Critiques and Limitations of the MVPF

Although the MVPF provides a powerful bridge between efficiency and equity, it too has conceptual and practical limitations that require careful interpretation.

1. **Dependence on Measured Outcomes:** MVPF estimates are only as good as the outcomes they include. Many benefits—improved health, civic engagement, reduced inequality—are difficult to quantify or monetize, leading to systematic underestimation of social value, especially for programs that enhance dignity, inclusion, or environmental sustainability.
2. **Choice of Welfare Weights:** The MVPF assumes that one dollar of gain to one person can be compared to another’s loss using social weights, but these weights are often implicit. Without explicit ethical assumptions about how society values gains to different income groups, MVPF analysis risks reproducing the same distributional blind spots as traditional cost–benefit analysis.
3. **Marginal vs. Systemic Effects:** Like the MCPF, the MVPF measures the effect of small, marginal changes to existing programs. Structural reforms—such as creating a guaranteed basic income or transforming the tax administration system—cannot be captured by

simple MVPF arithmetic because they alter the entire behavioural and institutional equilibrium.

4. **Time Horizon and Intergenerational Effects:** Many policies have delayed or intergenerational payoffs. Child benefits or postsecondary funding may yield returns decades later, while MVPF calculations typically discount future benefits heavily. Such discounting can bias policy evaluation against long-term or preventive investments.
5. **Administrative Capacity and Take-Up:** MVPF assumes that program benefits reach their intended recipients. In practice, non-filing, information barriers, and bureaucratic complexity can sharply reduce realized welfare gains. In Canada, for example, the MVPF of refundable tax credits depends critically on outreach and administrative design—reminding us that delivery is policy.
6. **Context and Counterfactual Bias:** MVPF estimates are highly context-specific: they depend on who participates, which taxes fund the spending, and what would have happened otherwise. Applying MVPF results from one setting to another without adjustment can mislead policymakers about true welfare impacts.

Despite these caveats, the MVPF remains a vital complement to the MCPF because it captures what society gets from the funds it raises. It transforms welfare economics from an abstract theory into a decision framework that explicitly values both efficiency and inclusion.

Beyond Efficiency: Toward a Broader Concept of Public Value

Together, the MCPF and MVPF extend welfare economics beyond efficiency alone. MCPF quantifies how much welfare is lost in raising funds; MVPF measures how much welfare is gained by spending them.

When used together, they form the analytical foundation for public value analysis—the idea that good fiscal policy maximizes the net social welfare created by government action. And when combined with material from the spending parts of the chapters of the text books, because MCPF and MVPF depend on behavioural and social structure, both incorporate distributional realities. Taxes that raise revenue from inelastic, high-income bases often have low MCPF; transfers targeted to low-income or equity-seeking groups often have high MVPF. Understanding this duality allows governments to pursue fiscal consolidation or expansion without neglecting inclusion.

But public value is not just a number. It reflects who counts in the measurement of costs and benefits, whose data are missing, and whose well-being is prioritized. The MVPF helps reveal those ethical and empirical choices. In this sense, the evolution from MCPF to MVPF represents not merely an analytical innovation but a normative shift in public finance—from efficiency to inclusion.

From Measuring Welfare to Designing Taxes

The MCPF and MVPF provide tools for assessing the welfare impact of particular policies—revealing which programs yield the greatest public value per dollar raised. But policymakers

rarely make decisions one program at a time. They must design entire tax systems capable of funding government activities efficiently, equitably, and sustainably. This requires moving from marginal evaluation to systemic design: from asking “Is this policy worthwhile?” to “What overall structure of taxation best supports our collective goals?”

Optimal-tax theory offers that framework. Building on the welfare-economics foundations developed throughout this chapter, it formalizes how governments can balance efficiency and equity across all taxes simultaneously. Early models, beginning with Frank Ramsey (1927), asked how to minimize efficiency costs when raising a fixed amount of revenue. Later work, most notably by James Mirrlees (1971), extended the approach to income taxation under real-world constraints of information and heterogeneity. Together, these frameworks form the core of modern tax design.

The next section develops these ideas step by step—showing how the same logic that underpins MCPF and MVPF can guide the design of fair, efficient, and inclusive tax systems.

The Logic of Optimal Taxation

Every government faces the same challenge: how to raise the revenues needed for collective well-being while preserving incentives to work, save, and invest. Theories of optimal taxation formalize this balancing act. They extend the welfare-economics framework developed in the previous sections to design tax systems that maximize social welfare subject to behavioural and informational constraints.

The Ramsey Rule—Minimizing Efficiency Costs

Frank Ramsey’s (1927) analysis asked a simple but enduring question: Given a fixed revenue requirement, how should taxes be spread across goods to minimize the total efficiency loss? His answer was the inverse-elasticity rule: tax rates should be inversely proportional to the price elasticity of demand for each good.

Goods with inelastic demand—such as necessities or land—should bear higher rates because they create smaller behavioural distortions; highly elastic goods should be taxed lightly. Formally, for commodities i and j :

$$\frac{t_i}{t_j} = \frac{(1 + t_j)\varepsilon_j}{(1 + t_i)\varepsilon_i}$$

where t is the tax rate and ε is the compensated price elasticity of demand.

In Figure 15.2, the more elastic demand curve shows a smaller deadweight-loss triangle for a given revenue required. The logic generalizes: raise revenue where behaviour changes least.

Critiques and limitations

Ramsey’s framework remains elegant but incomplete. It:

- Ignores distributional equity—who pays matters as much as efficiency.

- Assumes identical consumers and no administrative constraints.
- Treats consumption independently of income, overlooking fairness between people rather than goods.

These limitations led to the Mirrlees and Atkinson–Stiglitz frameworks that followed.

The Mirrlees Framework—Balance Equity and Efficiency

James Mirrlees (1971) reframed optimal taxation for income rather than commodities. Governments cannot observe individuals' true ability or effort; they observe only income. The challenge becomes designing a tax schedule $T(y)$ that maximizes social welfare subject to individuals choosing labour supply l to maximize after-tax income:

$$\underset{T(\cdot)}{Max} W = \sum_i U(c_i, l_i) \quad s.t. \quad c_i = y - T(y_i)$$

The result is the familiar trade-off: high marginal tax rates improve redistribution but reduce incentives to earn. The optimal marginal rate depends on the elasticity of labour supply and society's aversion to inequality.

Empirically, Mirrlees-type models imply:

- High top-end rates can be efficient if avoidance elasticities are modest.
- Negative or refundable taxes at the bottom (e.g., Canada Workers Benefit) improve welfare by encouraging participation.
- Smooth, progressive schedules dominate abrupt brackets that create large behavioural cliffs.

Integrating MCPF and MVPF—From Theory to Practice

The MCPF and MVPF introduced earlier can be interpreted as applied derivatives of Ramsey and Mirrlees logic.

- The MCPF identifies where marginal distortions are smallest (the efficiency side).
- The MVPF identifies where social welfare gains are largest (the equity/value side).

An optimal tax-and-transfer system equalizes the marginal social value of funds across programs and bases—that is, allocate resources until the MVPF of spending equals the MCPF of taxation.

When that equality holds, society cannot raise welfare further without reforming the underlying structure.

Contemporary Extensions and Critiques

Modern optimal-tax research adapts these ideas to the real world:

- Behavioural economics adds bounded rationality and framing effects (e.g., tax salience).

- Intersectional and gender analysis examines how elasticities differ across identity and care responsibilities.
- Administrative and evasion constraints recognize that feasible tax design must minimize avoidance and compliance costs.
- Environmental and wealth taxation extend Ramsey logic to intergenerational efficiency, where the “good” being taxed is pollution or idle capital rather than consumption.

In Canada, these insights underpin reforms such as income-tested credits, carbon pricing with rebates, and enhanced refundable benefits for families and low-income workers.

Policy Implications and Design Principles

Bringing the threads together yields several guiding principles for tax design:

1. Broad base, low rate. Minimize distortions by applying small wedges widely.
2. Vertical equity through progressivity. Balance fairness and efficiency via smooth marginal rates.
3. Neutrality across margins. Equalize effective tax rates on consumption, saving, and investment where possible.
4. Administrative simplicity. Design for compliance, transparency, and trust.
5. Inclusion and accessibility. Ensure all groups can claim credits and participate fully in the fiscal system.

An optimal system is not one that eliminates trade-offs but one that makes them explicit and deliberate.

In the News

Canada’s Wealth-Tax Debate and the Efficiency–Equity Frontier

In 2024, several parties revived calls for a federal wealth tax, citing rising inequality and the fiscal pressures of an aging population. Proposals ranged from a 1 percent annual levy on net worth above \$10 million to broader inheritance or capital-gains reforms. Proponents argued that wealth taxes score high on equity—targeting unearned gains among top households—while critics warned of capital flight and valuation challenges.

The debate reflects the Ramsey–Mirrlees trade-off in real time. If capital is highly mobile (elastic), the efficiency cost—its MCPF—could exceed one, suggesting limited revenue yield. But if the social return to greater equality and democratic legitimacy is large (high MVPF), overall welfare could rise even with some efficiency loss. The Parliamentary Budget Officer’s 2024 report estimated potential revenues between \$11 and \$20 billion annually, but also acknowledged behavioural responses that could halve those figures.

Sources: Parliamentary Budget Officer ([2024](#)); OECD ([2023](#)); Financial Post ([2024](#)).

Optimal Tax Design and Public Value

Optimal-tax theory brings together the tools introduced throughout this chapter. The Ramsey rule shows how to set tax rates to minimize efficiency loss across goods: tax bases with inelastic demand more heavily and those with elastic demand more lightly. The Mirrlees model extends this reasoning to the personal income tax, balancing equity and incentive effects to achieve the highest attainable level of social welfare.

In practice, these frameworks are not recipes but guiding principles. They help governments think systematically about trade-offs: between simplicity and precision, efficiency and fairness, national goals and provincial autonomy. Optimal design depends not only on elasticities and administrative capacity, but also on society's values and institutions.

Bringing MCPF and MVPF together within this framework allows policymakers to evaluate taxes and spending symmetrically. An efficient tax system minimizes the marginal cost of raising funds; a just and productive expenditure system maximizes the marginal value of using them. When both sides of the ledger are evaluated through these lenses, fiscal policy becomes a continuous process of aligning cost and benefit, efficiency and equity.

From Optimal Taxation to Fiscal Choice

These ideas also illuminate public debates over tax cuts, austerity, and debt financing.

- Tax cuts can raise welfare if they meaningfully reduce distortion (lower the MCPF) and if the lost revenue would otherwise have been used for low-value programs (low MVPF).
- Spending increases enhance welfare when they fund high-value programs (high MVPF) financed by relatively efficient taxes (low MCPF).
- Austerity—cutting both taxes and spending—can improve efficiency but risks reducing public value if essential services have MVPF well above one.
- Debt financing shifts the timing of both costs and benefits. Borrowing can raise welfare when it funds investments with $MVPF > 1$ that raise future revenue capacity, but it worsens welfare if it merely delays necessary tax adjustments.

In short, the combined MCPF–MVPF framework offers a disciplined, evidence-based way to move beyond slogans—“tax cuts pay for themselves” or “austerity is inevitable”—toward a transparent accounting of what society gains and loses from fiscal change.

Optimal taxation, in this modern sense, is not just about efficiency or fairness—it is about maximizing public value: using the nation's fiscal tools to expand opportunity, inclusion, and share

And into Part V

The analytical framework developed in this chapter—linking incidence, efficiency, equity, and optimal design—provides the foundation for Part V, which applies these principles to specific tax instruments: personal income, consumption, wealth, resource, and corporate taxation.

A side note for Selvia the details of which did not make it into the textbook chapter:

MVPF is defined as:

$$MVPF = \frac{\text{Welfare Gain to Recipients}}{\text{Net Fiscal Cost to Government}}.$$

That numerator—the welfare gain—is ideally the *sum of individuals' willingness to pay* for the policy (converted to utility terms). For cash transfer programs, for liquidity constrained households, when \$1 is transferred that notionally equals \$1 in WTP. In many in-kind programs (child care, health, housing), this value is difficult to observe and may require structural assumptions about preferences.

Critics argue that:

- Without a **structural model** of preferences and constraints, WTP cannot be measured precisely.
- Relying on observed behavioural responses (e.g., labour-supply elasticities) captures only part of welfare—ignoring non-market or quality-of-life gains.
- MVPF thus risks undervaluing programs with large unobserved or non-pecuniary benefits.

Hendren & Sprung-Keyser (2020) explicitly acknowledge this limitation. They treat MVPF as an **empirical approximation**, not a full welfare measure. In practice, they use *reduced-form estimates* of behavioural responses to infer WTP:

$$\begin{aligned} \text{Implied WTP} = & \text{Change in disposable income} \\ & + \text{Change in behaviour} \times \text{shadow value of time or consumption.} \end{aligned}$$

For example, in a child-care subsidy:

- If parents increase hours worked and pay more for care, that implies they value the subsidy at least as much as their out-of-pocket contribution.
- If utilization rises sharply when prices fall, revealed-preference methods can recover a lower-bound estimate of WTP.

So, rather than modelling WTP structurally, MVPF uses *empirical responses* to infer it—accepting this as an approximation. Empirical MVPF studies often use revealed-preference bounds from take-up and behavioural changes as proxies for WTP. This makes the MVPF an **approximate but transparent measure** of social return—strongest for monetary programs, less certain for goods with complex or unpriced benefits. As such, it complements rather than replaces full cost-benefit or structural welfare analysis. A summary of approximate approaches are below.

Bounding analysis (Hendren & Sprung-Keyser, 2020)

Reports a range (e.g., $MVPF \geq x$) when WTP is uncertain.

Partial-equilibrium revealed-preference methods

Uses observed take-up and price sensitivity to back out minimum WTP (as in public insurance or child-care demand).

Structural micro-simulation

Integrates MVPF with calibrated utility functions—data intensive but feasible for targeted programs.

Distributionally weighted MVPF

Places higher welfare weights on low-income groups, allowing use of fiscal incidence data instead of explicit WTP.